

Gender Inequality Index

Data Loading:

```
import numpy as np import pandas as pd import
matplotlib.pyplot as plt
data=pd.read_csv('/content/Gender_Inequality_Index.csv')
data.head()
```

...	Country	Human_development	GII	Rank	Maternal_mortality	Adolescent_birth_rate	Seats_parliament	F_
0	Switzerland	Very high	0.018	3.0	5.0	2.2	39.8	
1	Norway	Very high	0.016	2.0	2.0	2.3	45.0	
2	Iceland	Very high	0.043	8.0	4.0	5.4	47.6	
3	Hong Kong	Very high	NaN	NaN	NaN	1.6	NaN	
4	Australia	Very high	0.073	19.0	6.0	8.1	37.9	

Descriptive Statistics:

```
mean=data['GII'].mean() print(mean)
```

```
0.3443764705882353
```

```
median=data['GII'].median()
print(median)
```

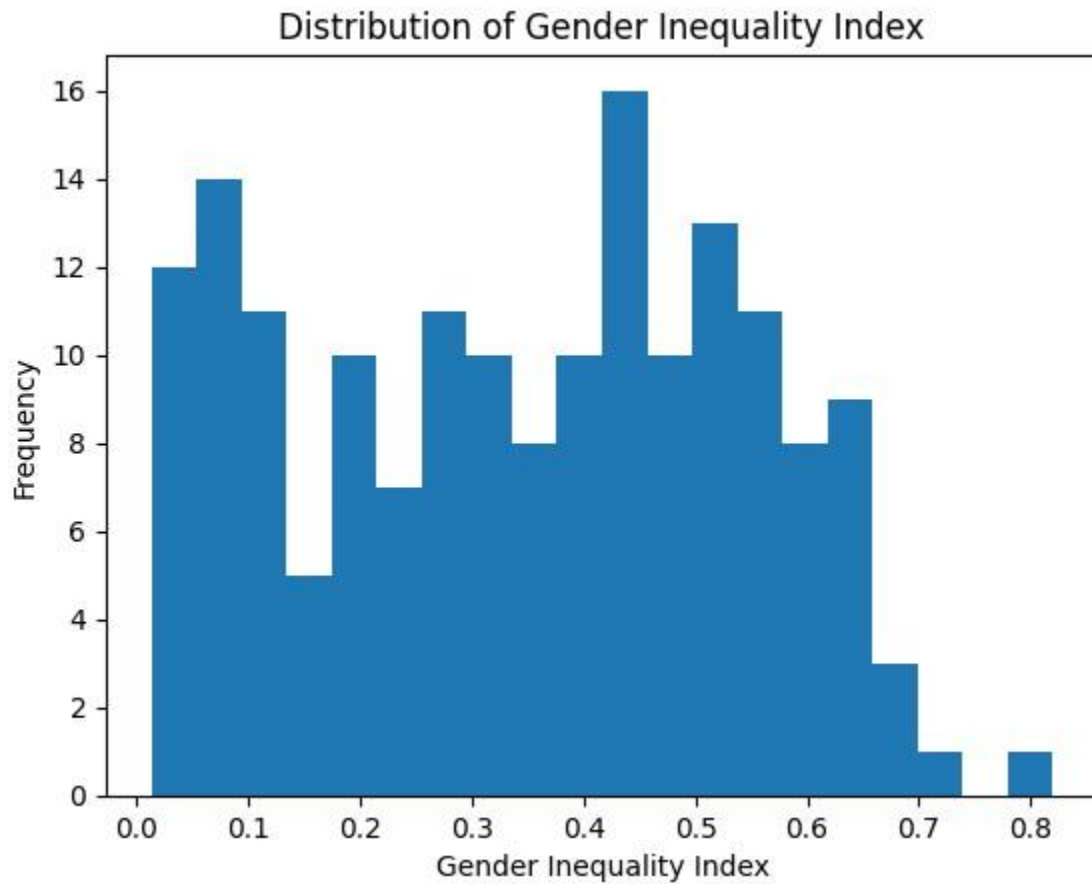
```
0.363
```

```
var=data['GII'].var()
print(var)
```

```
0.038850496484510964
```

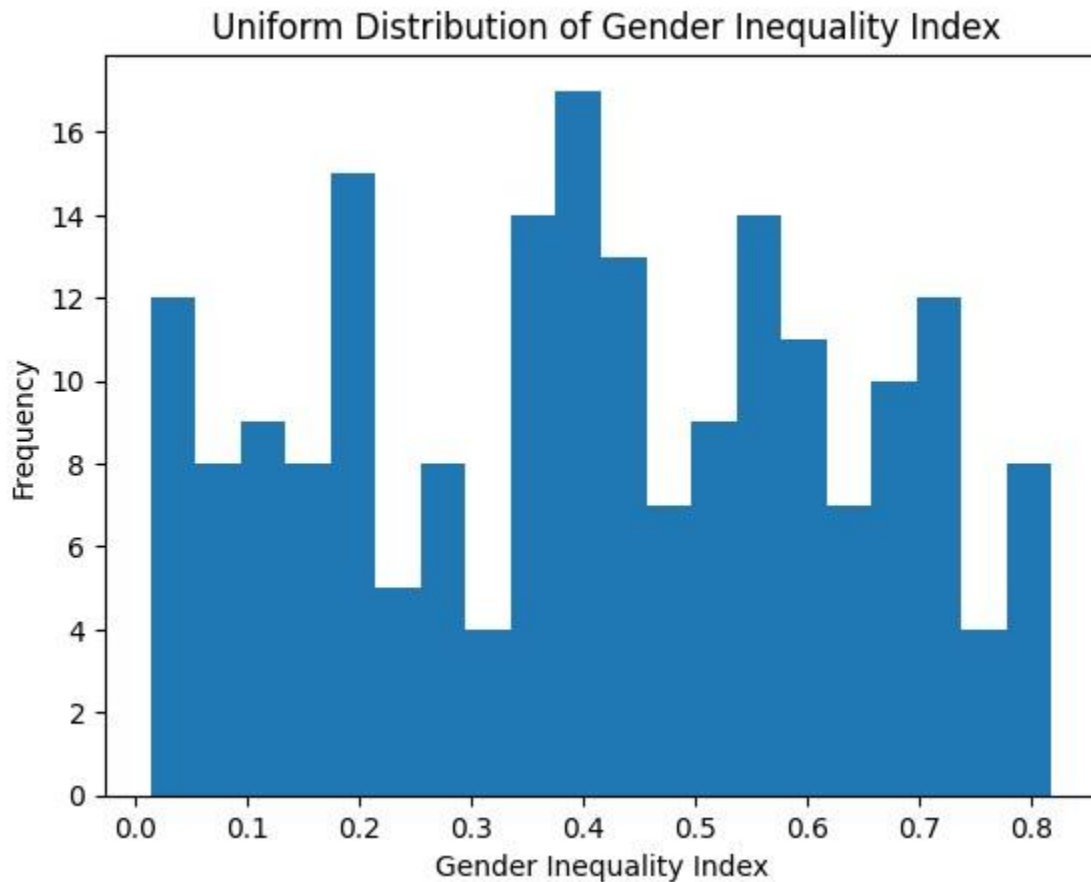
Distribution Analysis:

```
plt.hist(data['GII'],bins=20) plt.xlabel("Gender
Inequality Index") plt.ylabel("Frequency")
plt.title("Distribution of Gender Inequality Index")
plt.show()
```



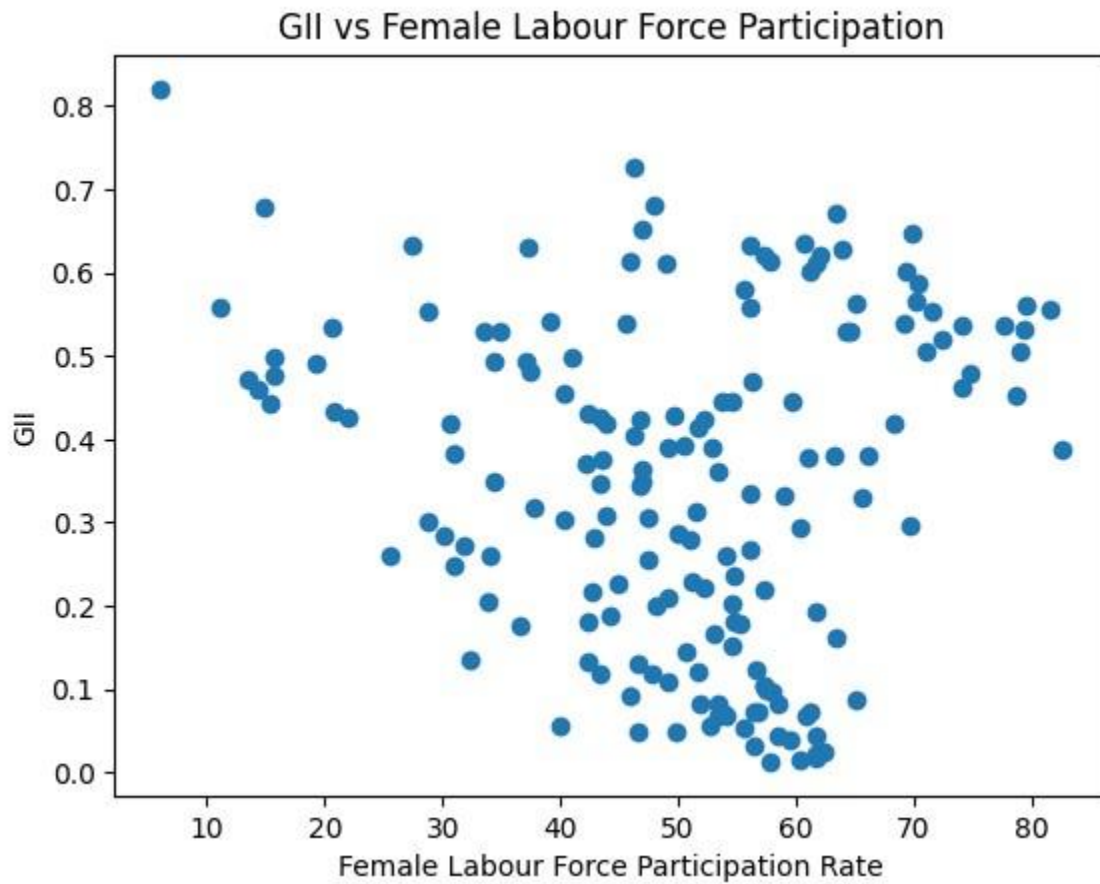
Uniform Distribution Comparison:

```
uniform_gii = np.random.uniform(    low=data['GII'].min(),  
high=data['GII'].max(),    size=len(data) )  
plt.hist(uniform_gii,bins=20) plt.xlabel("Gender Inequality  
Index") plt.ylabel("Frequency") plt.title(" Uniform  
Distribution of Gender Inequality Index") plt.show()
```



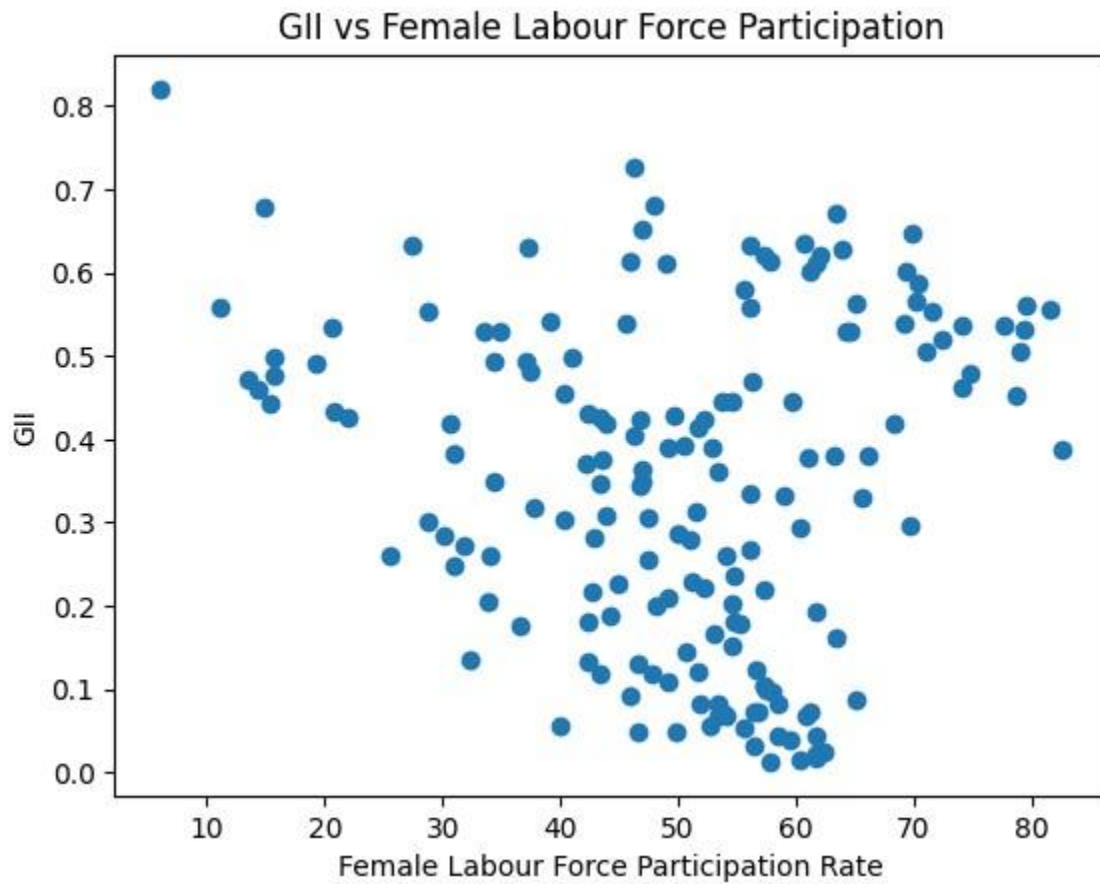
Correlation Analysis:

```
corr_flf=data['GII'].corr(data['F_Labour_force'])
corr_mm=data['GII'].corr(data['Maternal_mortality'])
print(corr_flf) print(corr_mm)
plt.scatter(data['F_Labour_force'], data['GII'])
plt.xlabel("Female Labour Force Participation Rate")
plt.ylabel("GII") plt.title("GII vs Female Labour
Force Participation") plt.show()
plt.scatter(data['Maternal_mortality'], data['GII'])
plt.xlabel("Maternal Mortality Ratio")
plt.ylabel("GII") plt.title("GII vs Maternal
Mortality Ratio") plt.show()
```



Visualization (Scatter Plots):

```
plt.scatter(data['F_Labour_force'], data['GII'])  
plt.xlabel("Female Labour Force Participation Rate")  
plt.ylabel("GII") plt.title("GII vs Female Labour  
Force Participation") plt.show()
```



```
plt.scatter(data['Maternal_mortality'],  
data['GII']) plt.xlabel("Maternal Mortality Ratio")  
plt.ylabel("GII") plt.title("GII vs Maternal  
Mortality Ratio") plt.show()
```

